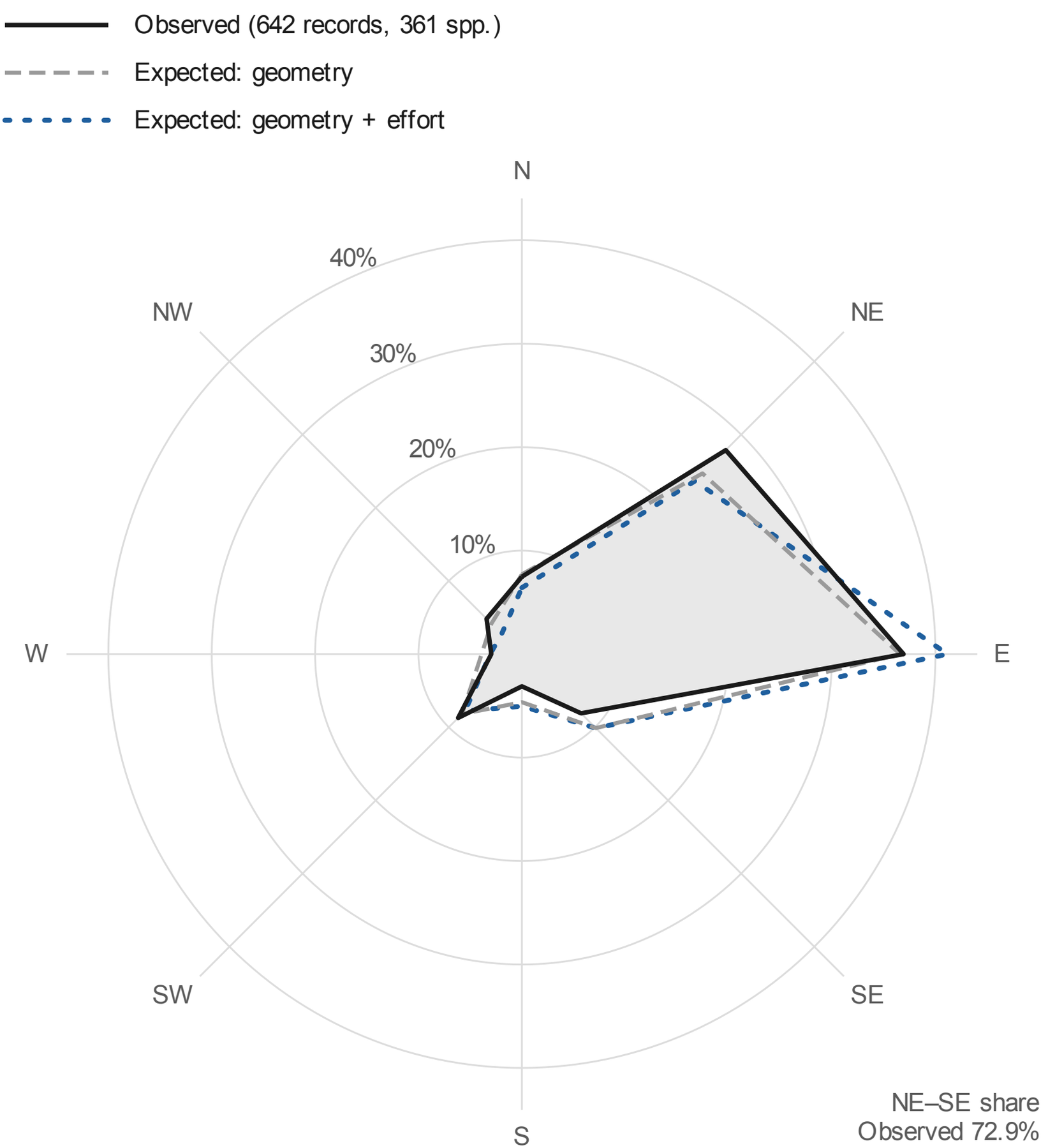Geometry alone reproduces the eastward skew